

Object Management Group

109 Highland Avenue
Needham, MA 02494
USA

Telephone: +1-781-444-0404
Facsimile: +1-781-444-0320
rfp@omg.org

Business Architecture Core Metamodel Request For Proposal

OMG Document: BMI/17-03-02

Letters of Intent due: *11 September 2017*
Submissions due: *19 February 2018*

Objective of this RFP

Business is changing at an increasing rate. At the same time, business complexity is increasing due to international expansion of markets, operational optimization and competitive pressures. Business leaders are challenged to meet this complexity and successfully adapt the business to remain competitive or realize competitive advantage.

There is widespread, sustained, and global proliferation of business architecture practices to support these management challenges. This is evidenced by the case studies presented at the OMG-sponsored Business Architecture Innovation Summits, increasing investment by tool vendors, growth in training course offerings, emergence of industry aligned business architecture reference models, and ongoing efforts by Open Group to align their business architecture approach to industry best practices.

However, much of this work is still based on diagrams and spreadsheets. IT modeling tools tend to be focused on design of information systems, obscuring

the definitions and relationships of the key elements of importance for understanding, analysis and strategy development by executives.

This RFP solicits proposals for a metamodel for core concepts of business architecture for business executives and other leaders to achieve the following:

- Support business architecture modeling as an integration of multiple perspectives to ensure that the business architecture is consistent and enables more effective management of business complexities.
- Provide views of the business architecture model that are at a level of abstraction appropriate for the understanding and analysis of business executives.
- Provide models based on a standard that enables the exchange of models between different modeling tools to avoid vendor lock-in and promote competition for quality and continued improvement of modeling tools.
- Support the development of consistent modeling techniques and methods to advance the business architecture discipline and support the training and certification of business architecture professionals.
- Facilitate the integration of multiple models of business, whether based on OMG standards or not.

For further details, see Section 6 of this document.

1 Introduction

1.1 Goals of OMG

The Object Management Group (OMG) is a software consortium with an international membership of vendors, developers, and end users. Established in 1989, its mission is to help computer users solve enterprise integration problems by supplying open, vendor-neutral portability, interoperability and reusability specifications based on Model Driven Architecture (MDA). MDA defines an approach to IT system specification that separates the specification of system functionality from the specification of the implementation of that functionality on a specific technology platform, and provides a set of guidelines for structuring specifications expressed as models. OMG has published many widely-used specifications such as UML [UML], BPMN [BPMN], MOF [MOF], XMI [XMI], DDS [DDS] and CORBA [CORBA], to name but a few significant ones.

1.2 Organization of this document

The remainder of this document is organized as follows:

Section 2 – *Architectural Context*. Background information on OMG's Model Driven Architecture.

Section 3 – *Adoption Process*. Background information on the OMG specification adoption process.

Section 4 – *Instructions for Submitters*. Explanation of how to make a submission to this RFP.

Section 5 – *General Requirements on Proposals*. Requirements and evaluation criteria that apply to all proposals submitted to OMG.

Section 6 – *Specific Requirements on Proposals*. Problem statement, scope of proposals sought, mandatory requirements, non-mandatory features, issues to be discussed, evaluation criteria, and timetable that apply specifically to this RFP.

Appendix A – References and Glossary Specific to this RFP

Appendix B – General References and Glossary

1.3 Conventions

The key words "**shall**", "**shall not**", "**should**", "**should not**", "**may**" and "**need not**" in this document should be interpreted as described in Part 2 of the ISO/IEC Directives [ISO2]. These ISO terms are compatible with the same terms in IETF RFC 2119 [RFC2119].

1.4 Contact Information

Questions related to OMG's technology adoption process and any questions about this RFP should be directed to rfp@omg.org.

OMG documents and information about the OMG in general can be obtained from the OMG's web site: <http://www.omg.org>. Templates for RFPs (like this document) and other standard OMG documents can be found on the Template Downloads Page: http://www.omg.org/technology/template_download.htm

2 Architectural Context

MDA provides a set of guidelines for structuring specifications expressed as models and the mappings between those models. The MDA initiative and the

standards that support it allow the same model, specifying business system or application functionality and behavior, to be realized on multiple platforms. MDA enables different applications to be integrated by explicitly relating their models; this facilitates integration and interoperability, and supports system evolution (deployment choices) as platform technologies change. The three primary goals of MDA are portability, interoperability and reusability.

Portability of any subsystem is relative to the subsystems on which it depends. The collection of subsystems that a given subsystem depends upon is often loosely called the *platform*, which supports that subsystem. Portability – and reusability – of such a subsystem is enabled if all the subsystems that it depends upon use standardized interfaces (APIs) and usage patterns.

MDA provides a pattern comprising a portable subsystem that is able to use any one of multiple specific implementations of a platform. This pattern is repeatedly usable in the specification of systems. The five important concepts related to this pattern are:

1. *Model* – A model is a representation of a part of the function, structure and/or behavior of an application or system. A representation is said to be formal when it is based on a language that has a well-defined form (“syntax”), meaning (“semantics”), and possibly rules of analysis, inference, or proof for its constructs. The syntax may be graphical or textual. The semantics might be defined, more or less formally, in terms of things observed in the world being described (e.g. message sends and replies, object states and state changes, etc.), or by translating higher-level language constructs into other constructs that have a well-defined meaning. The (non-mandatory) rules of inference define what unstated properties can be deduced from explicit statements in the model. In MDA, a representation that is not formal in this sense is not a model. Thus, a diagram with boxes and lines and arrows that is not supported by a definition of the meaning of a box, and the meaning of a line and of an arrow is not a model – it is just an informal diagram.
2. *Platform* – A set of subsystems/technologies that provide a coherent set of functionality through interfaces and specified usage patterns that any subsystem that depends on the platform can use without concern for the details of how the functionality provided by the platform is implemented.
3. *Platform Independent Model (PIM)* – A model of a subsystem that contains no information specific to the platform, or the technology that is used to realize it.

4. *Platform Specific Model (PSM)* – A model of a subsystem that includes information about the specific technology that is used in the realization of that subsystem on a specific platform, and hence possibly contains elements that are specific to the platform.
5. *Mapping* – Specification of a mechanism for transforming the elements of a model conforming to a particular metamodel into elements of another model that conforms to another (possibly the same) metamodel. A mapping may be expressed as associations, constraints, rules or templates with parameters that to be assigned during the mapping or other forms yet to be determined.

OMG adopts standard specifications of models that exploit the MDA pattern to facilitate portability, interoperability and reusability, either through *ab initio* development of standards or by reference to existing standards. Some examples of OMG adopted specifications are:

1. *Languages* – e.g. IDL for interface specification [IDL], UML for model specification [UML], BPMN for Business Process specification [BPMN], etc.
2. *Mappings* – e.g. Mapping of OMG IDL to specific implementation languages (CORBA PIM to Implementation Language PSMs), UML Profile for EDOC (PIM) to CCM (CORBA PSM) and EJB (Java PSM), CORBA (PSM) to COM (PSM) etc.
3. *Services* – e.g. Naming Service [NS], Transaction Service [OTS], Security Service [SEC], Trading Object Service [TOS] etc.
4. *Platforms* – e.g. CORBA [CORBA], DDS [DDS]
5. *Protocols* – e.g. GIOP/IOP [CORBA] (both structure and exchange protocol), DDS Interoperability Protocol [DDSI].
6. *Domain Specific Standards* – e.g. Model for Performance-Driven Government [MPG], Single Nucleotide Polymorphisms specification [SNP], TACSIT Controller Interface specification [TACSIT].

For an introduction to MDA, see [MDAa]. For a discourse on the details of MDA please refer to [MDAc]. To see an example of the application of MDA see [MDAb]. For general information on MDA, see [MDAd].

Object Management Architecture (OMA) is a distributed object computing platform architecture within MDA that is related to ISO's Reference Model of Open Distributed Processing RM-ODP [RM-ODP]. CORBA and any extensions to it are based on OMA. For information on OMA see [OMA].

3 Adoption Process

3.1 Introduction

OMG decides which specifications to adopt via votes of its Membership. The specifications selected should satisfy the architectural vision of MDA. OMG bases its decisions on both business and technical considerations. Once a specification is adopted by OMG, it is made available for use by both OMG members and non-members alike, at no charge.

This section 3 provides an extended summary of the RFP process. For more detailed information, see the *Policies and Procedures of the OMG Technical Process* [P&P], specifically Section 4.2, and the *OMG Hitchhiker's Guide* [Guide]. In case of any inconsistency between this document or the Hitchhiker's Guide and the Policies and Procedures, the P&P is always authoritative. All IPR-related matters are governed by OMG's *Intellectual Property Rights Policy* [IPR].

3.2 The Adoption Process in detail

3.2.1 Development and Issuance of RFP

RFPs, such as this one, are drafted by OMG Members who are interested in the adoption of an OMG specification in a particular area. The draft RFP is presented to the appropriate TF, discussed and refined, and when ready is recommended for issuance. If endorsed by the Architecture Board, the RFP may then be issued as an OMG RFP by a TC vote.

Under the terms of OMG's Intellectual Property Rights Policy [IPR], every RFP shall include a statement of the IPR Mode under which any resulting specification will be published. To achieve this, RFP authors choose one of the three allowable IPR modes specified in [IPR] and include it in the RFP – see section 6.10.

3.2.2 Letter of Intent (LOI)

Each OMG Member organization that intends to make a Submission in response to any RFP (including this one) shall submit a Letter of Intent (LOI) signed by an officer on or before the deadline specified in the RFP's timetable (see section 6.11). The LOI provides public notice that the organization may make a submission, but does not oblige it to do so.

3.2.3 Voter Registration

Any interested OMG Members, other than Trial, Press and Analyst members, may participate in Task Force voting related to this RFP. If the RFP timetable

includes a date for closing the voting list (see section 6.11), or if the Task Force separately decides to close the voting list, then only OMG Member that have registered by the given date and those that have made an Initial Submission may vote on Task Force motions related to this RFP.

Member organizations that have submitted an LOI are automatically registered to vote in the Task Force. Technical Committee votes are not affected by the Task Force voting list – all Contributing and Domain Members are eligible to vote in DTC polls relating to DTC RFPs, and all Contributing and Platform Members are eligible to vote in PTC polls on PTC RFPs.

3.2.4 Initial Submissions

Initial Submissions shall be made electronically on or before the Initial Submission deadline, which is specified in the RFP timetable (see section 6.11), or may later be adjusted by the Task Force. Submissions shall use the OMG specification template [TMPL], with the structure set out in section 4.9. Initial Submissions shall be written specifications capable of full evaluation, and not just a summary or outline. Submitters normally present their proposals to the Task Force at the first TF meeting after the submission deadline. Making a submission incurs obligations under OMG's IPR policy – see [IPR] for details.

An Initial Submission shall not be altered once the Initial Submission deadline has passed. The Task Force may choose to recommend an Initial Submission, unchanged, for adoption by OMG; however, instead Task Force members usually offer comments and feedback on the Initial Submissions, which submitters can address (if they choose) by making a later Revised Submission.

The goals of the Task Force's Submission evaluation are:

- Provide a fair and open process
- Facilitate critical review of the submissions by OMG Members
- Provide feedback to submitters enabling them to address concerns in their revised submissions
- Build consensus on acceptable solutions
- Enable voting members to make an informed selection decision

Submitters are expected to actively contribute to the evaluation process.

3.2.5 Revised Submissions

Revised Submissions are due by the specified deadline. Revised Submissions cannot be altered once their submission deadline has passed. Submitters again normally present their proposals at the next meeting of the TF after the deadline. If necessary, the Task Force may set a succession of Revised Submission deadlines. Submitters choose whether or not to make Revised Submissions - if they decide not to, their most recent Submission is carried forward, unless the Submitter explicitly withdraws from the RFP process.

The evaluation of Revised Submissions has the same goals listed above.

3.2.6 Selection Votes

When the Task Force's voters believe that they sufficiently understand the relative merits of the available Submissions, a vote is taken to recommend a submission to the Task Force's parent Technical Committee. The Architecture Board reviews the recommended Submission for MDA compliance and technical merit. Once the AB has endorsed it, members of the relevant TC vote on the recommended Submission by email. Successful completion of this vote moves the recommendation to OMG's Board of Directors (BoD).

3.2.7 Business Committee Questionnaire

Before the BoD makes its final decision on turning a Technical Committee recommendation into an OMG published specification, it asks its Business Committee to evaluate whether implementations of the specification will be publicly available. To do this, the Business Committee will send a Questionnaire [BCQ] to every OMG Member listed as a Submitter on the recommended Submission. Members that are not Submitters can also complete a Business Committee Questionnaire for the Submission if they choose.

If no organization commits to make use of the specification, then the BoD will typically not act on the recommendation to adopt it – so it is very important that submitters respond to the BCQ.

Once the Business Committee has received satisfactory BCQ responses, the Board takes the final publication vote. A Submission that has been adopted by the Board is termed an *Alpha Specification*.

At this point the RFP process is complete.

3.2.8 Finalization & Revision

Any specification adopted by OMG by any mechanism, whether RFP or otherwise, is subject to Finalization. A Finalization Task Force (FTF) is chartered by the TC that recommended the Specification; its task is to correct

any problems reported by early users of the published specification. The FTF first collaborates with OMG's Technical Editor to prepare a cleaned-up version of the Alpha Specification with submission-specific material removed. This is the Beta1 specification, and is made publicly available via OMG's web site. The FTF then works through the list of bug reports ("issues") reported by users of the Beta1 specification, to produce a Finalization Report and another Beta specification (usually Beta2), which is a candidate for Formal publication. Once endorsed by the AB and adopted by the relevant TC and BoD, this is published as the final, Formal Specification.

Long-term maintenance of OMG specifications is handled by a sequence of Revision Task Forces (RTFs), each one chartered to rectify any residual problems in the most-recently published specification version. For full details, see P&P section 4.4 [P&P].

4 Instructions for Submitters

4.1 OMG Membership

To submit to an RFP issued by the Platform Technology Committee an organization shall maintain either Platform or Contributing OMG Membership from the date of the initial submission deadline, while to submit to a Domain RFP an organization shall maintain either a Contributing or Domain membership.

4.2 Intellectual Property Rights

By making a Submission, an organization is deemed to have granted to OMG a perpetual, nonexclusive, irrevocable, royalty-free, paid up, worldwide license to copy and distribute the document and to modify the document and distribute copies of the modified version, and to allow others to do the same. Submitter(s) shall be the copyright owners of the text they submit, or have sufficient copyright and patent rights from the copyright owners to make the Submission under the terms of OMG's IPR Policy. Each Submitter shall disclose the identities of all copyright owners in its Submission.

Each OMG Member that makes a written Submission in response to this RFP shall identify patents containing Essential Claims that it believes will be infringed if that Submission is included in an OMG Formal Specification and implemented.

By making a written Submission to this RFP, an OMG Member also agrees to comply with the Patent Licensing terms set out in section 6.10.

This section 4.2 is neither a complete nor an authoritative statement of a submitter's IPR obligations – see [IPR] for the governing document for all OMG's IPR policies.

4.3 Submission Effort

An RFP submission may require significant effort in terms of document preparation, presentations to the issuing TF, and participation in the TF evaluation process. OMG is unable to reimburse submitters for any costs in conjunction with their submissions to this RFP.

4.4 Letter of Intent

Every organization intending to make a Submission against this RFP shall submit a Letter of Intent (LOI) signed by an officer on or before the deadline listed in section 6.11, or as later varied by the issuing Task Force.

The LOI should designate a single contact point within the submitting organization for receipt of all subsequent information regarding this RFP and the submission. The name of this contact will be made available to all OMG members. LOIs shall be sent by email, fax or paper mail to the “RFP Submissions Desk” at the OMG address shown on the first page of this RFP.

A suggested template for the Letter of Intent is available at <http://doc.omg.org/loi> [LOI].

4.5 Business Committee terms

This section contains the text of the Business Committee RFP attachment concerning commercial availability requirements placed on submissions. This attachment is available separately as OMG document omg/12-12-03.

4.5.1 Introduction

OMG wishes to encourage rapid commercial adoption of the specifications it publishes. To this end, there must be neither technical, legal nor commercial obstacles to their implementation. Freedom from the first is largely judged through technical review by the relevant OMG Technology Committees; the second two are the responsibility of the OMG Business Committee. The BC also looks for evidence of a commitment by a submitter to the commercial success of products based on the submission.

4.5.2 Business Committee evaluation criteria

4.5.2.1 *Viable to implement across platforms*

While it is understood that final candidate OMG submissions often combine technologies before they have all been implemented in one system, the Business Committee nevertheless wishes to see evidence that each major feature has been implemented, preferably more than once, and by separate organizations. Pre-product implementations are acceptable. Since use of OMG specifications should not be dependent on any one platform, cross-platform availability and interoperability of implementations should be also be demonstrated.

4.5.2.2 *Commercial availability*

In addition to demonstrating the existence of implementations of the specification, the submitter must also show that products based on the specification are commercially available, or will be within 12 months of the date when the specification was recommended for adoption by the appropriate Task Force. Proof of intent to ship product within 12 months might include:

- A public product announcement with a shipping date within the time limit.
- Demonstration of a prototype implementation and accompanying draft user documentation.

Alternatively, and at the Business Committee's discretion, submissions may be adopted where the submitter is not a commercial software provider, and therefore will not make implementations commercially available. However, in this case the BC will require concrete evidence of two or more independent implementations of the specification being used by end-user organizations as part of their businesses.

Regardless of which requirement is in use, the submitter must inform the OMG of completion of the implementations when commercially available.

4.5.2.3 *Access to Intellectual Property Rights*

OMG will not adopt a specification if OMG is aware of any submitter, member or third party which holds a patent, copyright or other intellectual property right (collectively referred to in this policy statement as "IPR") which might be infringed by implementation or recommendation of such specification, unless OMG believes that such IPR owner will grant an appropriate license to organizations (whether OMG members or not) which wish to make use of the specification. It is the goal of the OMG to make all of its technology available

with as few impediments and disincentives to adoption as possible, and therefore OMG strongly encourages the submission of technology as to which royalty-free licenses will be available.

The governing document for all intellectual property rights (“IPR”) policies of Object Management Group is the Intellectual Property Rights statement, available at: <http://doc.omg.org/ipr>. It should be consulted for the authoritative statement of the submitter's patent disclosure and licensing obligations.

4.5.2.4 Publication of the specification

Should the submission be adopted, the submitter must grant OMG (and its sublicensees) a worldwide, royalty-free license to edit, store, duplicate and distribute both the specification and works derived from it (such as revisions and teaching materials). This requirement applies only to the written specification, not to any implementation of it. Please consult the Intellectual Property Rights statement (<http://doc.omg.org/ipr>) for the authoritative statement of the submitter's copyright licensing obligations.

4.5.2.5 Continuing support

The submitter must show a commitment to continue supporting the technology underlying the specification after OMG adoption, for instance by showing the BC development plans for future revisions, enhancement or maintenance.

4.6 Responding to RFP items

4.6.1 Complete proposals

Submissions should propose full specifications for all of the relevant requirements detailed in Section 6 of this RFP. Submissions that do not present complete proposals may be at a disadvantage.

Submitters are encouraged to include any non-mandatory features listed in Section 6.

4.6.2 Additional specifications

Submissions may include additional specifications for items not covered by the RFP and which they believe to be necessary. Information on these additional items should be clearly distinguished. Submitters shall give a detailed rationale for why any such additional specifications should also be considered for adoption. Submitters should note that a TF is unlikely to consider additional items that are already on the roadmap of an OMG TF, since this would pre-empt the normal adoption process.

4.6.3 Alternative approaches

Submitters may provide alternative RFP item definitions, categorizations, and groupings so long as the rationale for doing so is clearly stated. Equally, submitters may provide alternative models for how items are provided if there are compelling technological reasons for a different approach.

4.7 Confidential and Proprietary Information

The OMG specification adoption process is an open process. Responses to this RFP become public documents of the OMG and are available to members and non-members alike for perusal. No confidential or proprietary information of any kind will be accepted in a submission to this RFP.

4.8 Proof of Concept

Submissions shall include a “proof of concept” statement, explaining how the submitted specifications have been demonstrated to be technically viable. The technical viability has to do with the state of development and maturity of the technology on which a submission is based. This is not the same as commercial availability. Proof of concept statements can contain any information deemed relevant by the submitter; for example:

“This specification has completed the design phase and is in the process of being prototyped.”

“An implementation of this specification has been in beta-test for 4 months.”

“A named product (with a specified customer base) is a realization of this specification.”

It is incumbent upon submitters to demonstrate the technical viability of their proposal to the satisfaction of the TF managing the evaluation process. OMG will favor proposals based on technology for which sufficient relevant experience has been gained.

4.9 Submission Format

4.9.1 General

- Submissions that are concise and easy to read will inevitably receive more consideration.
- Submitted documentation should be confined to that directly relevant to the items requested in the RFP.

- To the greatest extent possible, the submission should follow the document structure set out in "ISO/IEC Directives, Part 2 – Rules for the structure and drafting of International Standards" [ISO2]. An OMG specification template is available to make it easier to follow these guidelines.
- The key words "**shall**", "**shall not**", "**should**", "**should not**", "**may**" and "**need not**" shall be used as described in Part 2 of the ISO/IEC Directives [ISO2]. These ISO terms are compatible with the same terms in IETF RFC 2119 [RFC2119]. However, the RFC 2119 terms "**must**", "**must not**", "**optional**", "**required**", "**recommended**" and "**not recommended**" shall not be used (even though they are permitted under RFC2119).

4.9.2 Mandatory Outline

All submissions shall use the following structure, based on the OMG Specification template [TEMPL]:

Section 0 of the submission shall be used to provide all non-normative supporting material relevant to the evaluation of the proposed specification, including:

- The full name of the submission
- A complete list of all OMG Member(s) making the submission, with a named contact individual for each
- The acronym proposed for the specification (e.g. UML, CORBA)
- The name and OMG document number of the RFP to which this is a response
- The OMG document number of the main submission document
- Overview or guide to the material in the submission
- Statement of proof of concept (see 4.8)
- If the proposal does not satisfy any of the general requirements stated in Section 5, a detailed rationale explaining why
- Discussion of each of the "Issues To Be Discussed" identified in Section 6.
- An explanation of how the proposal satisfies the specific requirements and (if applicable) requests stated in Section 6.

- If adopting the submission requires making changes to already-adopted OMG specifications, include a list of those changes in a clearly-labelled subsection in Section 0. Identify exactly which version(s) of which OMG specification(s) shall be amended, and include the list of precise wording changes that shall be made to that specification.

Section 1 and subsequent sections of the submission shall contain the normative specification that the Submitter(s) is/are proposing for adoption by OMG, including:

- Scope of the proposed specification
- Overall design rationale
- Conformance criteria for implementations of the proposed specification, clearly stating the features that all conformant implementations shall support, and any features that implementations may support, but which are not mandatory.
- A list of the normative references that are used by the proposed specification
- A list of terms that are used in the proposed specification, with their definitions
- A list of any special symbols that are used in the proposed specification, together with their significance
- The proposed specification itself

Section 0 will be deleted from any specification that OMG adopts and publishes. Therefore Section 0 of the submission shall contain no normative material (other than any instructions to change existing specifications; ensuring that these are implemented is the responsibility of the FTF that finalizes the specification, before it deletes section 0). Any non-normative material outside section 0 shall be explicitly identified.

The main submission document and any models or other machine-interpretable files accompanying it shall be listed in an inventory file conforming to the inventory template [INVENT].

The submission shall include a copyright waiver in a form acceptable to OMG. One acceptable form is:

“Each of the entities listed above: (i) grants to the Object Management Group, Inc. (OMG) a nonexclusive, royalty-free, paid up, worldwide license to copy

and distribute this document and to modify this document and distribute copies of the modified version, and (ii) grants to each member of the OMG a nonexclusive, royalty-free, paid up, worldwide license to make up to fifty (50) copies of this document for internal review purposes only and not for distribution, and (iii) has agreed that no person shall be deemed to have infringed the copyright in the included material of any such copyright holder by reason of having used any OMG specification that may be based hereon or having conformed any computer software to such specification.”

Other forms of copyright waiver may only be used if approved by OMG legal counsel beforehand.

4.10 How to Submit

Submitters should send an electronic version of their submission to the *RFP Submissions Desk* (rfp@omg.org) at OMG Headquarters by 5:00 PM U.S. Eastern Standard Time (22:00 GMT) on the day of the Initial and Revised Submission deadlines. Acceptable formats are Adobe FrameMaker source, ISO/IEC 26300:2006 (OpenDoc 1.1), OASIS DocBook 4.x (or later) and ISO/IEC 29500:2008 (OOXML, .docx).

Submitters should ensure that they receive confirmation of receipt of their submission.

5 General Requirements on Proposals

5.1 Requirements

5.1.1 Use of modeling languages

Submitters are encouraged to express models using OMG modeling languages such as UML, MOF, CWM and SPEM (subject to any further constraints on the types of the models and modeling technologies specified in Section 6 of this RFP). Submissions containing models expressed using OMG modeling languages shall be accompanied by an OMG XMI [XMI] representation of the models (including a machine-readable copy). A best effort should be made to provide an OMG XMI representation even in those cases where models are expressed via non-OMG modeling languages.

5.1.2 PIMs & PSMs

Section 6 of this RFP specifies whether PIM(s), PSM(s), or both are being solicited. If proposals specify a PIM and corresponding PSM(s), then the rules specifying the mapping(s) between the PIM and PSM(s) shall either be identified by reference to a standard mapping or specified in the proposal. In order to allow

possible inconsistencies in a proposal to be resolved later, proposals shall identify whether it's the mapping technique or the resulting PSM(s) that shall be considered normative.

5.1.3 Complete submissions

Proposals shall be *precise* and *functionally complete*. Any relevant assumptions and context necessary to implement the specification shall be provided.

5.1.4 Reuse

Proposals shall *reuse* existing OMG and other standard specifications in preference to defining new models to specify similar functionality.

5.1.5 Changes to existing specifications

Each proposal shall justify and fully specify any *changes or extensions* to existing OMG specifications necessitated by adopting that proposal. In general, OMG favors proposals that are *upwards compatible* with existing standards and that minimize changes and extensions to existing specifications.

5.1.6 Minimalism

Proposals shall factor out functionality that could be used in different contexts and specify their models, interfaces, etc. separately. Such *minimalism* fosters re-use and avoids functional duplication.

5.1.7 Independence

Proposals shall use or depend on other specifications only where it is actually necessary. While re-use of existing specifications to avoid duplication will be encouraged, proposals should avoid gratuitous use.

5.1.8 Compatibility

Proposals shall be *compatible* with and *usable* with existing specifications from OMG and other standards bodies, as appropriate. Separate specifications offering distinct functionality should be usable together where it makes sense to do so.

5.1.9 Implementation flexibility

Proposals shall preserve maximum *implementation flexibility*. Implementation descriptions should not be included and proposals shall not constrain implementations any more than is necessary to promote interoperability.

5.1.10 Encapsulation

Proposals shall allow *independent implementations* that are *substitutable* and *interoperable*. An implementation should be replaceable by an alternative implementation without requiring changes to any client.

5.1.11 Security

In order to demonstrate that the specification proposed in response to this RFP can be made secure in environments that require security, answers to the following questions shall be provided:

- What, if any, security-sensitive elements are introduced by the proposal?
- Which accesses to security-sensitive elements should be subject to security policy control?
- Does the proposed service or facility need to be security aware?
- What default policies (e.g., for authentication, audit, authorization, message protection etc.) should be applied to the security sensitive elements introduced by the proposal? Of what security considerations should the implementers of your proposal be aware?

The OMG has adopted several specifications, which cover different aspects of security and provide useful resources in formulating responses. [SEC] [RAD].

5.1.12 Internationalization

Proposals shall specify the degree of internationalization support that they provide. The degrees of support are as follows:

- a) Uncategorized: Internationalization has not been considered.
- b) Specific to <region name>: The proposal supports the customs of the specified region only, and is not guaranteed to support the customs of any other region. Any fault or error caused by requesting the services outside of a context in which the customs of the specified region are being consistently followed is the responsibility of the requester.
- c) Specific to <multiple region names>: The proposal supports the customs of the specified regions only, and is not guaranteed to support the customs of any other regions. Any fault or error caused by requesting the services outside of a context in which the customs of at least one of the specified regions are being consistently followed is the responsibility of the requester.
- d) Explicitly not specific to <region(s) name>: The proposal does not support the customs of the specified region(s). Any fault or error caused by requesting

the services in a context in which the customs of the specified region(s) are being followed is the responsibility of the requester.

5.2 Evaluation criteria

Although the OMG adopts model-based specifications and not implementations of those specifications, the technical viability of implementations will be taken into account during the evaluation process. The following criteria will be used:

5.2.1 Performance

Potential implementation trade-offs for performance will be considered.

5.2.2 Portability

The ease of implementation on a variety of systems and software platforms will be considered.

5.2.3 Securability

The answer to questions in section 5.1.11 shall be taken into consideration to ascertain that an implementation of the proposal is securable in an environment requiring security.

5.2.4 Conformance: Inspectability and Testability

The adequacy of proposed specifications for the purposes of conformance inspection and testing will be considered. Specifications should provide sufficient constraints on interfaces and implementation characteristics to ensure that conformance can be unambiguously assessed through both manual inspection and automated testing.

5.2.5 Standardized Metadata

Where proposals incorporate metadata specifications, OMG standard XMI metadata [XMI] representations should be provided.

6 Specific Requirements on Proposals

6.1 Problem Statement

The business architecture efforts noted in the Objectives section of this RFP have not produced a standard metamodel for business architecture, nor sought to create a standard modeling method for business architecture. In addition, they have not produced a modeling scope for business architecture relative to other

modeling disciplines used in business analysis and strategy, nor have they indicated how business architecture models would integrate with other models used to describe business intent, structure and operations.

The objective of this RFP is to solicit proposals resulting in the adoption of a standard metamodel specification for the practice of business architecture. Such a standard specification will incent vendors to create tools to support the practice of business architecture, promote the interchange of business architecture models between tools, and improve the consistency of practice of business architecture, thereby enhancing the repeatability of business analysis results based on business architecture.

In this RFP, the term “business” is intended to mean any organized effort directed at realizing a social, governance or profit objective.

Businesses are complex and dynamic efforts and can be modelled in many different ways, depending on the analysis question being asked. The OMG has many adopted specifications allowing the modeling of information technology, for example. Likewise, it has several adopted specifications for modeling process flows, case management and decision management. It also has an adopted specification for modeling strategies.

Business decision makers, strategists and analysts also need to model the creation of value, the perception of value by stakeholders, the impact of transformation on business structure, the role of business information (as distinct from data) in operational and strategic decision making and the identification of business capabilities needed by the business and implemented in multiple places both inside and outside of the business.

Each of the adopted specifications mentioned above, and each of the issues that are not addressed by these specifications can be said to define a perspective on a business being modelled. Perspectives isolate the features of a business that are relevant to addressing particular types of business problems. Perspectives also simplify the task of the modeler and avoid the distraction of irrelevant detail when such models and analysis results derived from them are being presented.

However, because perspectives are views on an underlying organization and its behavior, they cannot be fully independent of each other and they cannot contradict each other. Consequently, modeling specifications shall allow for these perspectives to be aligned so that concepts are not duplicated and contradictions can be detected and resolved.

6.2 Scope of Proposals Sought

This RFP defines scope in terms of a collection of perspectives, exemplified in the following text. These perspectives are given names to suggest their content and business purpose. The perspective names are terms believed to be commonly used by business people, but they do not have generally agreed definitions. The expectation of the RFP is that submitters will have experience in business modeling and provision of tools to business users, and will use this experience to guide how they scope the content of each perspective.

There is a growing belief among business people that an abstract model of a business that describes what a business does without describing how it does it, is a key perspective that can be used as a point of alignment among multiple other perspectives. Such a model has come to be called a capability model. Submissions shall include a metamodel for a business capability perspective, defined as follows:

- Capability – as a network of concepts that describe what a business of a selected type does to create value without describing how, when, or with what resources¹.

While business capability is a key integrating notion for business models, it is only one essential business concept. Submitters should address in their submissions, additional perspectives needed to facilitate the business modeling objectives set out in section 6.1.

Figure 1 shows a summary of common business perspectives, with the capability perspective as the focal point, surrounded by other business perspectives. Capability is a core concept of business, and in business architecture models, it enables other perspectives to be linked together in a way that permits multi-perspective analysis and strategy formulation activities to be carried out and enabling “360 degree” views of current and future businesses.

These perspectives are not intended as an exhaustive list, but are examples of business perspectives that should be incorporated into a submission. Submitters should propose additional perspectives that are useful in strategy formulation and diagnosis of business issues. Submitters are not constrained to align concepts and groups of concepts in their submissions with these perspectives and categories. Submissions should incorporate a sufficient number of perspectives to allow the creation of adequate business architecture models.

¹ See e.g. Homann, Ulrich, “A Business-Oriented Foundation for Service Orientation”, <https://msdn.microsoft.com/en-us/library/aa479368.aspx>, 2006

In addition, there is a generalized category at the bottom of the figure called “processes, decisions, and events” that can be expanded upon to align related disciplines. This bottom category is meant to address linkages to business perspectives with existing standard metamodels, such as BPMN.

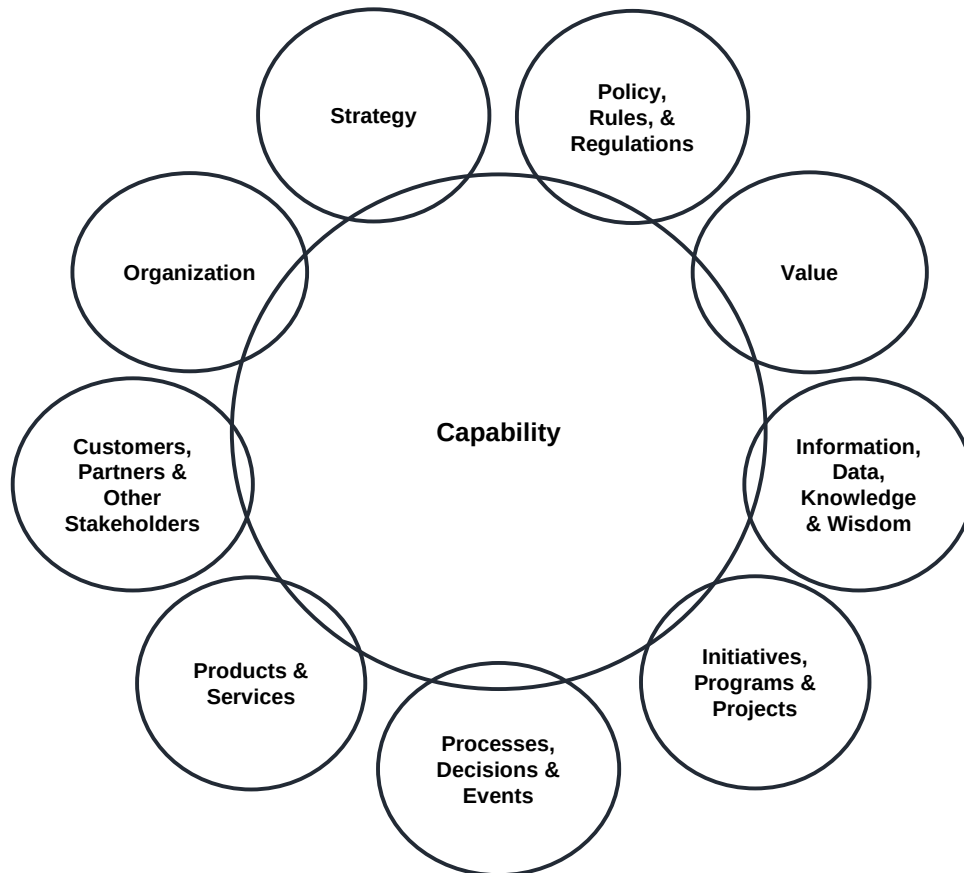


Figure 1: Capability Oriented Business Perspective

It is important to note that the view shown in figure 1 does not represent relationships with, associations to, or alignment to IT architecture. Such an alignment is important to both business and IT and is contemplated as the subject of a separate RFP. Submitters shall respond to this RFP with suggestions about such alignment (see section 6.7).

The term “touchpoint” is used in this RFP to refer to interactions between perspectives, including perspectives that are expressed in models external to the business architecture model. Touchpoint interactions are often expressed by business people as rules or constraints. Submitters shall identify such touchpoints when they are believed to be material to the successful employment of models based on their submission.

Touchpoints are used to address two RFP goals;

1. provide a business context for submission teams seeking to communicate how the submitted standard would be used;
2. keep the RFP from being overly prescriptive regarding touchpoints (i.e. by specifying in this RFP, explicit and required relationships among business perspectives) when this task is better left to submission teams.

Examples of business interactions that could be considered as touchpoints are given below.

- Enable a business to determine strategic objectives impacting certain capabilities
- Articulate capabilities necessary to enable the delivery of customer value
- Determine capabilities impacted by a given initiative, program, or project
- Understand how business units share capabilities with other business units
- Determine capabilities packaged within or needed to enable a product or service
- Show information required as an input to one or more capabilities
- Define the capabilities required to deliver a given product or service to a customer
- Determine capabilities linked to a given business process

As used by business people, touchpoints are often expressed as rules that involve two or more business concepts. While simple touchpoints are able to be implemented as associations (binary relations), using associations as the mechanism for touchpoints is problematic for several reasons: 1) touchpoint rules involving 3 or more elements require the modeler to use a pattern for implementing n-ary relations with binary associations; 2) the touchpoint rule may apply complex restrictions on several of the elements participating in the rule and binary associations can only place constraints on the endpoint types and cardinalities of that association; 3) direct association of elements in two different models joins the models in a semantic union, and this union can contain contradictions and other undesirable consequences.

In Sections 6.5 and 6.6 of this RFP, submitters are asked to propose a mechanism for specifying touchpoints.

6.3 Relationship to other OMG Specifications and activities

6.3.1 Relationship to OMG specifications

See section 6.5.2 for specific requirements regarding accommodation of and integration with other OMG Specifications.

See Appendix A.1 for references to the OMG specifications mentioned in this RFP.

6.3.1.1 OMG specifications that shall be considered

The Value Delivery Model and Language [VDML] specification provides a standard modeling language for the operation of an enterprise, with particular focus on the creation and exchange of value. VDML is valuable as an elaboration of business architecture to capture additional details about business operations and value creation, including the capture of metrics. Submissions shall consider the VDML specification.

The Business Motivation Model [BMM] is often used to capture strategy and mission. It shall be considered by submitters intending to include the strategy perspective in their submissions.

The Business Process Model And Notation™ (BPMN™) [BPMN] is used to capture both abstract and detailed business processes.

See sections 6.5, 6.6 and 6.7 for additional requirements regarding VDML, BPMN and BMM.

Submissions are shall provide a [MOF] metamodel (see Appendix A.2).

6.3.1.2 OMG specifications that should be considered

The following list of OMG specifications is intended to be suggestive to submitters and is not intended to be normative or exhaustive. Submitters shall refer to the public list of OMG specifications to determine relevant specifications.

The Case Management Model and Notation [CMMN] and Decision Model Notation [DMN] are also used to describe business behaviors.

Semantics of Business Vocabulary and Rules™ (SBVR™) [SBVR] is a general facility for modeling business terminology, definitions and rules.

The Unified Profile for the Department of Defense Architecture Framework (DoDAF) and the Ministry of Defence Architecture Framework (MODAF)

(UPDM) defines a UML profile for modeling mission-oriented command and control structures and behaviors. UPDM is being superseded by the Unified Architecture Framework Profile [UAF].

The Structured Metrics Metamodel [SMM] is an OMG specification for representing measurement information related to any model structured information. SMM is relevant when a submission intends to provide a facility for modeling measurements. See section 6.5 for requirements involving the SMM specification.

The Unified Modeling Language (UML) [UML]- The objective of UML is to provide system architects, software engineers, and software developers with tools for analysis, design, and implementation of software-based systems as well as for modeling business and similar processes. UML models have also been used to define and describe the structure and lifecycle of information.

XMI [XMI] is an XML model interchange (serialization) format for interchanging models created using the submitted metamodel.

6.3.2 Relationship to other OMG Documents and work in progress

Semantic Information Modeling for Federation [SIMF] is a work in progress to establish a mechanism for federation of information. This work in progress is relevant to the touchpoint mechanism described in this RFP.

6.4 Related non-OMG Activities, Documents and Standards

6.4.1 The Open Group

[Capability Guide] A Guide published by The Open Group and authored by the Architecture Forum Business Architecture Work Stream - <https://www2.opengroup.org/ogsys/catalog/g161>.

[Value Stream Guide] A Guide published by The Open Group and authored by the Architecture Forum Business Architecture Work Stream - <https://www2.opengroup.org/ogsys/catalog/G170>.

[Archimate 3.0] A specification published by The Open Group Archimate Forum - <https://www2.opengroup.org/ogsys/catalog/I162M>

6.4.2 Business Architecture Guild

[BIZBOK] A Guide to the Business Architecture Body of Knowledge® - (BIZBOK® Guide) published by the Business Architecture Guild – <http://www.businessarchitectureguild.org>

6.5 Mandatory Requirements

6.5.1 General Requirements

Submissions shall provide one MOF metamodel for entities, attributes and relationships.

6.5.2 Specific Requirements

6.5.2.1 *Capability perspective*

The metamodel submission shall include the capability perspective. Capability² is defined as a framework of concepts that describe what a business of a selected type does to create value without describing how, when, or with what resources³.

6.5.2.2 *Glossary*

For each peripheral perspective⁴ in a submission, the submitters shall include a glossary of terms used in the submitted metamodel as labels or in constraints, with a business-oriented definition of each such term.

Each term in said glossary shall be supported by one or more references to non-proprietary business standards or business publications that demonstrate that the term and its definition are substantially used in the business community.

6.5.2.3 *Touchpoint specification mechanism*

Each submission containing one or more peripheral perspectives shall show a mechanism for specifying for the touchpoints between each peripheral perspective metamodel with the capability perspective metamodel. This mechanism shall be able to represent, for example, that the production of value for a stakeholder requires certain capabilities. In addition, the mechanism shall be usable for touchpoints between peripheral perspectives.

² The term “capability” has been used in many different ways. For example, in mission planning, a capability is defined as a specific set of resources needed to carry out a specific action in a mission. This RFP seeks a more abstract definition of capability that is descriptive of types of activity and may include general requirements needed to carry out the activity as annotations.

³ See e.g. Homann, Ulrich, “A Business-Oriented Foundation for Service Orientation”, <https://msdn.microsoft.com/en-us/library/aa479368.aspx>, 2006

⁴ In this document, the term “peripheral perspective” shall refer to any business perspective in a submission other than the capability perspective

6.5.2.4 *Distinguishing Business Architecture elements from similar-named elements in external models*

Existing OMG standards such as UPDM/UAF present metamodel elements and relationships and define terms and concepts that can be confused with terms and concepts used in business architecture. Submitters should identify possible confusions and provide definitions for business architecture terms and concepts that distinguish them from similar terms and concepts in other standards. Submitters are encouraged to identify where a business architecture term or concept is related to a similar term or concept in another OMG standard and to describe the nature of the relationship.

6.6 Non-mandatory features

6.6.1 Recommended submission perspectives

6.6.1.1 *Value perspective*

The submitted metamodel should include the value perspective, incorporating modeling how value is perceived by stakeholders⁵ and modeling how value is created in parts or increments⁶.

6.6.1.2 *Information perspective*

The submitted metamodel should include the information perspective. This perspective shall allow modeling of information as used in strategic and operational decision making and stakeholder reporting. Every business produces information for these purposes, and submitted metamodels shall show how such information results from capabilities specific to information production and handling.

Business people view information differently from IT people. Business people are interested in the provenance and reliability of information, and information is not always rendered as data. Business people use information to make decisions and the quality of decisions is dependent on the quality of information. Business information often represents a truth that is vague, contains contradictions, and relative to a particular context. Submissions of data schema models may be deemed as not satisfying the needs of the business user of information.

⁵ See Pigneur, Y and Osterwalder, A. "How to Design and Test Business Models and Value Propositions", <http://www.slideshare.net/ypigneur/value-proposition-design-47698911>

⁶ See Porter, M., Competitive Advantage, Creating and Sustaining Superior Performance, 1985 or Whittle, R. and Myrick, C., Enterprise Business Architecture: The Formal Link between Strategy and Results, 2004.

6.6.1.3 *Organization perspective*

The submitted metamodel should include the organization perspective. Submissions should be capable of modeling command and control relationships, and should be capable of modeling communication, collaboration, budget disbursement relationships and other permanent and temporary organization structures.

6.6.1.4 *Integration of non-mandatory perspectives with Capability perspective*

Submissions that elect to include perspectives listed in this RFP section include a touchpoint specifications defined by the touchpoint specification mechanism with the submitted capability perspective. These specifications shall model the “touchpoints” the submitter believes to be relevant and useful in the practice of business architecture.

In addition to the core business architecture perspectives, submitters are encouraged to submit alignment mechanisms that permit alignment with existing OMG and other standard metamodels for business modeling. The term “alignment” is intended to mean that models created using submissions to this RFP shall be able to be linked with other models created according to existing OMG standards and should be able to be linked to models used in business strategy and analysis and not governed by any OMG standard. This mechanism should be, but is not required to be the same as the touchpoint specification mechanism. Submitters should provide an alignment mechanism for additional modeling standards that they believe to be materially important to the successful use of models based on their submission.

6.6.2 **Additional perspectives**

Submissions should include additional perspectives, whether included in Figure 1 above or not. Such metamodel submissions are to meet the general requirements as stated above. However, submitters are discouraged from submitting metamodels that cover topics and categories of marginal interest to business people or are hybrids of business and technology⁷.

Submitters should describe in general terms how submitted metamodels could be used in developing IT architectures to support business activities and structures described by their submitted metamodels. This description is intended as a rationale that the proposed metamodels would have utility in an IT architecture process. A separate RFP is planned to request metamodels or other mechanisms to support alignment of business architecture and IT architecture.

⁷ It is anticipated that future RFP(s) will address the alignment of business architecture models with models for technology systems.

6.6.3 Alignment with business processes defined using BPMN

It is common practice to use BPMN to define abstract business processes and to use methods similar to Model Driven Architecture (MDA) to elaborate these abstract models into detailed business process models. Such abstract business processes are often quite similar to value streams/value chains. Submitters should recognize this practice in their responses and describe how models created using their proposed metamodel can be aligned with abstract and detailed business process models created from the BPMN standard.

6.6.4 Using quantitative measures from external models

Existing OMG standards for business process modeling (BPMN) and business operational modeling (VDML) are typically used for business analysis projects that include quantitative modeling and/or simulation. Business architecture, on the other hand, is used to represent value creation streams and business structures in business analysis projects that do not include quantitative modeling or simulation. Simple color assignments (green, yellow, red) and simple ranks (1-5) are used in business architecture to convey summary information highlighting elements of the business architecture model where executive attention is being recommended.

Because a quantitative analysis effort is often needed to produce the summary diagnostic information using in a business architecture model, submissions should recognize the possibility that models based on standards such as BPMN and VDML will be used to capture and analyze quantitative information. This recognition leads to two additional considerations for RFP submitters:

1. Submissions to this RFP should include a simple facility for annotating business architecture model elements, or should include a recommendation for an adjunct metamodel standard for metrics (such as SMM).
2. Submissions to this RFP should describe how metrics from a BPMN or VDML model can be associated with a business architecture model and used to establish the colors/ranks for business architecture model elements.

If a submitter elects to include metrics modeling, the OMG Structured Metrics Metamodel [SMM] shall be supported.

6.7 Issues to be discussed

6.7.1 Touchpoints

This RFP requests submitters to provide one or more mechanisms for specifying touchpoints between business architecture models and models that have been created based on other metamodels, whether such metamodels are OMG

standards or not. A simple mechanism might consist of incorporation of other metamodels and the establishment of specific relationships between entities in the distinct metamodels. However, experience has shown that this mechanism is difficult for tool vendors to implement, fragile, and restricts the applicability of the mechanism in practice thus reducing its value. Submissions shall discuss how the proposed touchpoint mechanism addresses:

- Fragility of linkages between models created at different times and with different tools;
- Establishment of linkages between elements in separate models where such elements are partially semantically overlapping;
- Establishment of linkages between elements in separate models that restrict the visibility of features and other elements related to a linked element, as seen from the element that is the source of the link.

6.7.2 Relevance to business concepts

For each perspective in a submission, submitters shall discuss how this perspective is used by business people, preferably by reference to an established body of practice described in generally available business oriented publications. In addition, submitters shall discuss how the entities and relationships in applicable representational diagrams and tables can be captured in models instantiated from the proposed metamodel.

6.7.3 Relationship to existing OMG specifications

Submitters should specify that an existing OMG standard metamodel is to be used for modeling of one or more perspectives. In this case, the submitted touchpoint mechanism shall be used to link the OMG standard metamodel to the required capability perspective metamodel elements and to other relevant perspectives metamodel elements. Submissions shall discuss the rationale for reusing an existing standard metamodel, including any particular interpretations placed on the standard metamodel when used in conjunction with business architecture.

If a metamodel submitted for a peripheral perspective overlaps with or differs from an existing OMG standard, submissions shall discuss any overlaps with or differences from the existing OMG standard. Such discussions shall provide a rationale for the overlap or difference, and explain how the existing standard would be modified or re-interpreted.

Submitters shall discuss mappings of the business architecture concepts represented in their submission to business relevant OMG specifications BMM,

BPMN and VDML. Submissions should discuss mappings to OMG specifications such as UML, CMMN, DMN and UPDM/UAF. Additional information about and requirements on these discussions can be found in sections 6.5 and 6.6 of this RFP.

These discussions will be considered during submission evaluation. They will not be part of the proposed normative specification.

6.7.4 Linkage with enterprise IT architecture

This RFP does not require submitters to propose touchpoints to OMG metamodel standards commonly used for enterprise IT architecture; it is anticipated that such touchpoints will be the subject of a future RFP. Because of the importance of aligning business architecture and IT architecture, submitters shall discuss the important types of touchpoints they see as relevant to this alignment (e.g. the alignment of software systems with the capabilities they support). Additionally, submitters shall discuss the suitability of their touchpoint mechanism proposals for aligning business architecture metamodel elements with enterprise IT architecture elements.

6.8 Evaluation Criteria

Submissions shall meet the mandatory requirements set forth in section 6.5 of this RFP.

Submissions will be evaluated based on the degree to which they meet the optional requirements; in other words, a submission that includes more optional content will be graded higher. However, optional content that has been specifically encouraged by this RFP will be evaluated higher than other optional content.

Submissions will be evaluated on the linkages between concepts in the metamodels and on the adequacy of these linkages to express important business relationships.

Submitted mechanisms for model alignment will be evaluated on semantic completeness and precision. For an example of such a mechanism and related evaluation criteria, see the “The Distributed Ontology, Modeling and Specification Language (DOL)” OMG adopted specification [DOL]. Because an objective of this RFP is to facilitate the integration of multiple types of commonly used business models by way of a business architecture metamodel, the submitted mechanism will be evaluated by its ability to support model integration.

Submitters are required to provide information in their submissions showing that the concepts in their submissions are based on concepts found in business

publications and practices. Submissions will be judged by completeness in justifying metamodel concepts by reference to such publications and practices. Submissions will also be judged by the commonality and quality of referenced publications and practices.

Submissions will be evaluated on the quality and completeness of the discussions on alignment with business relevant OMG standard metamodels such as VDML, BMM, BPMN, DMN, CMMN and UPDM/UAF.

6.9 Other information unique to this RFP

None

6.10 IPR Mode

Every OMG Member that makes any written Submission in response to this RFP shall provide the Non-Assertion Covenant found in Appendix A of the OMG IPR Policy [IPR].

6.11 RFP Timetable

The timetable for this RFP is given below. Note that the TF or its parent TC may, in certain circumstances, extend deadlines while the RFP is running, or may elect to have more than one Revised Submission step. The latest timetable can always be found at the *OMG Work In Progress* page at <http://www.omg.org/schedules> under the item identified by the name of this RFP.

Event or Activity	Date
<i>Letter of Intent (LOI) deadline</i>	<i>11 September 2017</i>
<i>Initial Submission deadline</i>	<i>19 February 2018</i>
<i>Voter registration closes</i>	<i>27 November 2017</i>
<i>Initial Submission presentations</i>	<i>19 March 2018</i>
<i>Revised Submission deadline</i>	
<i>Revised Submission presentations</i>	

Appendix A References & Glossary Specific to this RFP

A.1 References Specific to this RFP

[BMM] The Business Motivation Model (<http://www.omg.org/spec/BMM/1.3>)

[BPMN] The Business Process Model And Notation™ (BPMN™):
<http://www.omg.org/spec/BPMN/2.0.2/>.

[CMMN] Case Management Model and Notation
<http://www.omg.org/spec/CMMN/Current>

[DMN] Decision Model Notation <http://www.omg.org/spec/DMN/Current>

[DOL] The Distributed Ontology, Modeling and Specification Language (DOL)
OMG adopted specification (currently OMG document ptc-16-02-37.pdf).

[SBVR] Semantics of Business Vocabulary and Rules™ (SBVR™)
<http://www.omg.org/spec/SBVR/1.3>.

[SIMF] Semantic Information Modeling for Federation (SIMF)
http://www.omgwiki.org/architecture-ecosystem/doku.php?id=semantic_information_modeling_for_federation_rfp

[SMM] Structured Metrics Meta-Model (SMM)
<http://www.omg.org/spec/SMM/Current>

[UAF] Unified Architecture Framework Profile (UAF)
<http://www.omg.org/spec/UAF/1.0/Beta1/>.

[UML] Unified Modeling Language <http://www.omg.org/spec/UML/2.5>.

[UPDM] The Unified Profile for the Department of Defense Architecture Framework (DoDAF) and the Ministry of Defense Architecture Framework (MODAF) (UPDM) <http://www.omg.org/spec/UPDM/2.1>.

[VDML] The Value Delivery Modeling Language (VDML)
<http://www.omg.org/spec/VDML/1.0/>.

[XMI] XML Metadata Interchange. The current specification is:
<http://www.omg.org/spec/XMI/2.5.1>.

A.2 Glossary Specific to this RFP

Touchpoint - The term “touchpoint” is used in this RFP to refer to interactions between perspectives, including perspectives that may be expressed in models external to the business architecture model. Touchpoint interactions are often expressed by business people as rules or constraints.

Capability - Capability is defined as a framework of concepts that describe what a business of a selected type does to create value without describing how, when, or with what resources.

Appendix B General Reference and Glossary

B.1 General References

The following documents are referenced in this document:

[BCQ] OMG Board of Directors Business Committee Questionnaire,
<http://doc.omg.org/bcq>

[CCM] CORBA Core Components Specification
<http://www.omg.org/spec/CCM/>

[CORBA] Common Object Request Broker Architecture (CORBA)
<http://www.omg.org/spec/CORBA/>

[CORP] UML Profile for CORBA,
<http://www.omg.org/spec/CORP>

[CWM] Common Warehouse Metamodel Specification
<http://www.omg.org/spec/CWM>

[EDOC] UML Profile for EDOC Specification
<http://www.omg.org/spec/EDOC/>

[Guide] The OMG Hitchhiker's Guide
<http://doc.omg.org/hh>

[IDL] Interface Definition Language Specification
<http://www.omg.org/spec/IDL35>

[INVENT] Inventory of Files for a Submission/Revision/Finalization
<http://doc.omg.org/inventory>

[IPR] IPR Policy

<http://doc.omg.org/ipr>

[ISO2] ISO/IEC Directives, Part 2 – Rules for the structure and drafting of International Standards

<http://isotc.iso.org/livelink/livelink?func=ll&objId=4230456>

[LOI] OMG RFP Letter of Intent template

<http://doc.omg.org/loi>

[MDAa] OMG Architecture Board, "Model Driven Architecture - A Technical Perspective"

<http://www.omg.org/mda/papers.htm>

[MDAb] Developing in OMG's Model Driven Architecture (MDA)

<http://www.omg.org/mda/papers.htm>

[MDAc] MDA Guide

<http://www.omg.org/docs/omg/03-06-01.pdf>

[MDAd] MDA "The Architecture of Choice for a Changing World"

<http://www.omg.org/mda>

[MOF] Meta Object Facility Specification

<http://www.omg.org/spec/MOF/>

[NS] Naming Service

<http://www.omg.org/spec/NAM>

[OMA] Object Management Architecture

<http://www.omg.org/oma/>

[OTS] Transaction Service

<http://www.omg.org/spec/OTS>

[P&P] Policies and Procedures of the OMG Technical Process

<http://doc.omg.org/pp>

[RAD] Resource Access Decision Facility

<http://www.omg.org/spec/RAD>

[ISO2] ISO/IEC Directives, Part 2 – Rules for the structure and drafting of International Standards

<http://isotc.iso.org/livelink/livelink?func=ll&objId=4230456>

[RM-ODP]
ISO/IEC 10746

[SEC] CORBA Security Service
<http://www.omg.org/spec/SEC>

[TEMPL] Specification Template
<http://doc.omg.org/submission-template>

[TOS] Trading Object Service
<http://www.omg.org/spec/TRADE>

[UML] Unified Modeling Language Specification,
<http://www.omg.org/spec/UML>

[XMI] XML Metadata Interchange Specification,
<http://www.omg.org/spec/XMI>

B.2 General Glossary

Architecture Board (AB) - The OMG plenary that is responsible for ensuring the technical merit and MDA-compliance of RFPs and their submissions.

Board of Directors (BoD) - The OMG body that is responsible for adopting technology.

Common Object Request Broker Architecture (CORBA) - An OMG distributed computing platform specification that is independent of implementation languages.

Common Warehouse Metamodel (CWM) - An OMG specification for data repository integration.

CORBA Component Model (CCM) - An OMG specification for an implementation language independent distributed component model.

Interface Definition Language (IDL) - An OMG and ISO standard language for specifying interfaces and associated data structures.

Letter of Intent (LOI) - A letter submitted to the OMG BoD's Business Committee signed by an officer of an organization signifying its intent to respond to the RFP and confirming the organization's willingness to comply with OMG's terms and conditions, and commercial availability requirements.

Mapping - Specification of a mechanism for transforming the elements of a model conforming to a particular metamodel into elements of another model that conforms to another (possibly the same) metamodel.

Metadata - Data that represents models. For example, a UML model; a CORBA object model expressed in IDL; and a relational database schema expressed using CWM.

Metamodel - A model of models.

Meta Object Facility (MOF) - An OMG standard, closely related to UML, that enables metadata management and language definition.

Model - A formal specification of the function, structure and/or behavior of an application or system.

Model Driven Architecture (MDA) - An approach to IT system specification that separates the specification of functionality from the specification of the implementation of that functionality on a specific technology platform.

Normative – Provisions to which an implementation shall conform to in order to claim compliance with the standard (as opposed to non-normative or informative material, included only to assist in understanding the standard).

Normative Reference – References to documents that contain provisions to which an implementation shall conform to in order to claim compliance with the standard.

Platform - A set of subsystems/technologies that provide a coherent set of functionality through interfaces and specified usage patterns that any subsystem that depends on the platform can use without concern for the details of how the functionality provided by the platform is implemented.

Platform Independent Model (PIM) - A model of a subsystem that contains no information specific to the platform, or the technology that is used to realize it.

Platform Specific Model (PSM) - A model of a subsystem that includes information about the specific technology that is used in the realization of it on a specific platform, and hence possibly contains elements that are specific to the platform.

Request for Information (RFI) - A general request to industry, academia, and any other interested parties to submit information about a particular technology area to one of the OMG's Technology Committee subgroups.

Request for Proposal (RFP) - A document requesting OMG members to submit proposals to an OMG Technology Committee.

Task Force (TF) - The OMG Technology Committee subgroup responsible for issuing a RFP and evaluating submission(s).

Technology Committee (TC) - The body responsible for recommending technologies for adoption to the BoD. There are two TCs in OMG – the *Platform TC* (PTC) focuses on IT and modeling infrastructure related standards; while the *Domain TC* (DTC) focuses on domain specific standards.

Unified Modeling Language (UML) - An OMG standard language for specifying the structure and behavior of systems. The standard defines an abstract syntax and a graphical concrete syntax.

UML Profile - A standardized set of extensions and constraints that tailors UML to particular use.

XML Metadata Interchange (XMI) - An OMG standard that facilitates interchange of models via XML documents.